

RainMind: Complete Mathematical Formula Handbook

Grid-based Regime-Aware NWP Rainfall Post-Processing, MoE, EMOS, QM, Verification and Continual Learning

200 formulas organized layer-by-layer for a TensorFlow / PyTorch prototype

Scope. This handbook expands the formulas in the supplied formula note and organizes them into a prototype-ready mathematical toolbox. It includes deterministic bias correction, regime probabilities, spatial/graph layers, sparse Mixture-of-Experts (MoE), LoRA/FiLM conditioning, Quantile Mapping (QM), EMOS/CSG-EMOS, BMA-style probabilistic post-processing, uncertainty, extremes, and the SIH-style verification metrics.

Important implementation note. The 200 formulas are a library of candidate components, not 180 equations that must be trained simultaneously. For an MVP, select a small, auditable subset; add advanced layers only after ablation testing shows incremental value. In particular, verify rainfall distribution assumptions and zero-rain handling before choosing an EMOS distribution.

Source integration. The supplied note already contains the core router, transition, memory, LoRA, FiLM, MLP, correction, and complexity formulas. Those are retained and expanded here. The supplied note's core equations include the router input, softmax routing, transition magnitude, grid-specific prior, MoE correction, and LoRA/FiLM conditioning equations. The handbook also adds standard verification, QM, EMOS/CSG-EMOS, CRPS, spatial verification, threshold, extreme-value, and continual-learning formulas.

Recommended formula flow

1. Grid and geometry -> spatial indexing and lead time.
2. Preprocessing -> normalization and physically meaningful transforms.
3. Regime discovery/classification -> probabilistic regime state.
4. Spatial-temporal perception -> neighbourhood and transport features.
5. Transition field -> probability change, temporal smoothing and regime memory.
6. MoE -> router, Top-K selection and expert correction.
7. Statistical post-processing baselines -> scaling, QM, EMOS/CSG-EMOS and BMA.
8. Predictive distribution -> P10/P50/P90, exceedance probabilities and uncertainty.
9. Verification -> RMSE, ETS, CSI, POD, FAR, FSS, ROC/AUC, Brier, CRPS.
10. Continual learning -> error memory, drift detection and controlled online update.

1. Grid, geometry, indexing and lead time

1. **Latitude spacing.** Grid latitude increment.

$$\Delta\phi = \phi_{i+1} - \phi_i \quad (1)$$

2. **Longitude spacing.** Grid longitude increment.

$$\Delta\lambda = \lambda_{j+1} - \lambda_j \quad (2)$$

3. **Approximate north-south cell size.** Physical cell height; use radians internally.

$$\Delta y \approx R_{\oplus} \Delta\phi \frac{\pi}{180} \quad (3)$$

4. **Approximate east-west cell size.** Physical cell width varies with latitude.

$$\Delta x \approx R_{\oplus} \cos(\phi) \Delta\lambda \frac{\pi}{180} \quad (4)$$

5. **Cell area.** Area weighting for grid aggregation.

$$A_{cell}(\phi) \approx R_{\oplus}^2 \left(\frac{\pi}{180} \right)^2 \Delta\phi \Delta\lambda \cos \phi \quad (5)$$

6. **Grid count.** For 135 x 129, $N_{grid} = 17,415$ points.

$$N_{grid} = N_{\phi} N_{\lambda} \quad (6)$$

7. Grid index. Flatten 2-D grid to 1-D index; choose zero- or one-based convention consistently.

$$g = iN_\lambda + j \quad (7)$$

8. Lead-time conversion. Forecast lead time in hours.

$$L_h = \frac{t_{valid} - t_{init}}{3600 \text{ s}} \quad (8)$$

9. Great-circle distance. Distance between grid centers.

$$d_{ij} = 2R_\oplus \arcsin \left(\sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos \phi_i \cos \phi_j \sin^2 \frac{\Delta\lambda}{2}} \right) \quad (9)$$

10. Bearing. Direction from grid i to grid j.

$$\theta_{ij} = \text{atan2}(\sin \Delta\lambda \cos \phi_j, \cos \phi_i \sin \phi_j - \sin \phi_i \cos \phi_j \cos \Delta\lambda) \quad (10)$$

2. Preprocessing, normalization and data transforms

11. Arithmetic mean. Baseline normalization and diagnostics.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (11)$$

12. Weighted mean. Use cell-area or reliability weights.

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i} \quad (12)$$

13. Sample variance. Spread estimate.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (13)$$

14. Standard deviation. Feature scaling and ensemble spread.

$$s = \sqrt{\frac{1}{n-1} \sum_i (x_i - \bar{x})^2} \quad (14)$$

15. Z-score. Standardization for neural and statistical models.

$$z_i = \frac{x_i - \mu}{\sigma} \quad (15)$$

16. Min-max scaling. Scale to [0,1].

$$x'_i = \frac{x_i - x_{min}}{x_{max} - x_{min}} \quad (16)$$

17. Robust scaling. Less sensitive to rainfall extremes.

$$x'_i = \frac{x_i - Q_{0.50}}{Q_{0.75} - Q_{0.25}} \quad (17)$$

18. Log1p rainfall transform. Compress right-skewed precipitation.

$$x' = \log(1 + x) \quad (18)$$

19. Inverse Log1p. Return to rainfall units.

$$x = \exp(x') - 1 \quad (19)$$

20. Box-Cox transform. Variance-stabilizing transform for positive data.

$$x_\lambda = \frac{x^\lambda - 1}{\lambda} \quad (\lambda \neq 0), \quad x_0 = \log x \quad (20)$$

3. Error, bias and descriptive statistics

21. Mean error / bias. Signed systematic error.

$$\text{Bias} = \frac{1}{n} \sum_i (\hat{y}_i - y_i) \quad (21)$$

22. Absolute error. Pointwise magnitude of error.

$$e_i = |\hat{y}_i - y_i| \quad (22)$$

23. Mean absolute error. Robust deterministic error metric.

$$MAE = \frac{1}{n} \sum_i |\hat{y}_i - y_i| \quad (23)$$

24. Mean squared error. Squared-error loss and verification.

$$MSE = \frac{1}{n} \sum_i (\hat{y}_i - y_i)^2 \quad (24)$$

25. Root mean squared error. SIH deterministic verification.

$$RMSE = \sqrt{\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2} \quad (25)$$

26. Normalized RMSE. Scale-independent comparison.

$$NRMSE = \frac{RMSE}{y_{max} - y_{min}} \quad (26)$$

27. Mean absolute percentage error. Use cautiously when observed rainfall is near zero.

$$MAPE = \frac{100}{n} \sum_i \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (27)$$

28. Symmetric MAPE. More stable than MAPE around small values.

$$sMAPE = \frac{100}{n} \sum_i \frac{|\hat{y}_i - y_i|}{(|\hat{y}_i| + |y_i|)/2} \quad (28)$$

29. Correlation coefficient. Association, not direct calibration.

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \quad (29)$$

30. Coefficient of determination. Variance explained relative to climatology mean.

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (30)$$

4. Probability distributions and density functions

31. Bernoulli PMF. Binary rain-event occurrence.

$$P(Y = y) = p^y (1 - p)^{1-y}, \quad y \in \{0, 1\} \quad (31)$$

32. Bernoulli mean. Event probability.

$$\mathbb{E}[Y] = p \quad (32)$$

33. Bernoulli variance. Occurrence uncertainty.

$$\text{Var}(Y) = p(1 - p) \quad (33)$$

34. Normal PDF. EMOS for approximately continuous variables.

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right] \quad (34)$$

35. Normal CDF. Probability of rainfall below a threshold under Normal EMOS.

$$F(y) = \Phi\left(\frac{y-\mu}{\sigma}\right) \quad (35)$$

36. Lognormal PDF. Positive, right-skewed rainfall alternative.

$$f(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left[-\frac{(\ln y - \mu)^2}{2\sigma^2}\right], \quad y > 0 \quad (36)$$

37. Gamma PDF. Positive rainfall intensity model.

$$f(y) = \frac{y^{k-1} e^{-y/\theta}}{\theta^k \Gamma(k)}, \quad y > 0 \quad (37)$$

38. Gamma moments. Convert gamma shape/scale to mean/variance.

$$\mathbb{E}[Y] = k\theta, \quad \text{Var}(Y) = k\theta^2 \quad (38)$$

39. Gamma parameters from moments. Used in CSG-EMOS parameterization.

$$k = \frac{\mu^2}{\sigma^2}, \quad \theta = \frac{\sigma^2}{\mu} \quad (39)$$

40. Gumbel PDF. Extreme-value baseline for sensitivity analysis.

$$f(y) = \frac{1}{\beta} \exp[-z - e^{-z}], \quad z = \frac{y-\mu}{\beta} \quad (40)$$

5. Regime discovery and classification

41. GMM mixture density. Probabilistic regime discovery.

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k) \quad (41)$$

42. GMM responsibility. Soft regime membership.

$$\gamma_{ik} = P(z_i = k \mid x_i) = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)} \quad (42)$$

43. GMM log-likelihood. Train/select GMM parameters.

$$\ell(\Theta) = \sum_i \log \left[\sum_k \pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k) \right] \quad (43)$$

44. GMM BIC. Select cluster count/model complexity.

$$BIC = -2\ell_{max} + p \log n \quad (44)$$

45. GMM AIC. Alternative model-selection criterion.

$$AIC = -2\ell_{max} + 2p \quad (45)$$

46. HDBSCAN core distance. Density notion for HDBSCAN.

$$\text{core}_k(x) = \text{k-distance}(x) \quad (46)$$

47. Mutual reachability distance. HDBSCAN graph edge distance.

$$d_{mreach-k}(a, b) = \max\{\text{core}_k(a), \text{core}_k(b), d(a, b)\} \quad (47)$$

48. Silhouette coefficient. Cluster separation/compactness.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (48)$$

49. Multiclass cross-entropy. Train supervised regime classifier.

$$\mathcal{L}_{CE} = -\frac{1}{n} \sum_i \sum_k y_{ik} \log p_{ik} \quad (49)$$

50. Multiclass Brier score. Calibrate regime probabilities.

$$BS_{multi} = \frac{1}{n} \sum_i \sum_k (p_{ik} - y_{ik})^2 \quad (50)$$

6. Softmax, calibration, entropy and regime confidence

51. Softmax. Regime or MoE probabilities.

$$p_k = \frac{e^{z_k/T}}{\sum_j e^{z_j/T}} \quad (51)$$

52. Stable softmax. Numerically stable implementation.

$$p_k = \frac{e^{z_k - z_{max}}}{\sum_j e^{z_j - z_{max}}} \quad (52)$$

53. Temperature scaling. Post-hoc probability calibration.

$$p_k(T) = \text{softmax}(z/T)_k \quad (53)$$

54. Binary sigmoid. Binary event head.

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (54)$$

55. Log-odds. Probability transform.

$$\text{logit}(p) = \log \frac{p}{1-p} \quad (55)$$

56. Regime entropy. Low = concentrated regime; high = ambiguous/transitioning.

$$H(p) = -\sum_k p_k \log p_k \quad (56)$$

57. Normalized entropy. Comparable uncertainty for different K.

$$H_{norm} = \frac{-\sum_k p_k \log p_k}{\log K} \quad (57)$$

58. Entropy-based confidence. Simple regime confidence index.

$$C = 1 - H_{norm} \quad (58)$$

59. Expected calibration error. Probability calibration diagnostic.

$$ECE = \sum_{b=1}^B \frac{|S_b|}{n} |\text{acc}(S_b) - \text{conf}(S_b)| \quad (59)$$

60. Maximum calibration error. Worst-bin calibration error.

$$MCE = \max_b |\text{acc}(S_b) - \text{conf}(S_b)| \quad (60)$$

7. Spatial perception, convolution and graph learning

61. 2-D convolution. Local spatial feature extraction.

$$Y_{i,j,c} = \sum_{u,v,d} K_{u,v,d,c} X_{i+u,j+v,d} + b_c \quad (61)$$

62. Dilated convolution. Larger receptive field without large kernels.

$$Y_{i,j,c} = \sum_{u,v,d} K_{u,v,d,c} X_{i+ru,j+rv,d} + b_c \quad (62)$$

63. Average pooling. Spatial aggregation.

$$Y_{i,j,c} = \frac{1}{|\mathcal{N}|} \sum_{(u,v) \in \mathcal{N}} X_{i+u,j+v,c} \quad (63)$$

64. Max pooling. Dominant-feature aggregation.

$$Y_{i,j,c} = \max_{(u,v) \in \mathcal{N}} X_{i+u,j+v,c} \quad (64)$$

65. Adjacency from distance. Static spatial graph.

$$A_{ij} = \exp\left(-\frac{d_{ij}^2}{2\sigma_d^2}\right) \mathbf{1}[d_{ij} \leq r_0] \quad (65)$$

66. Wind alignment. Downwind influence score.

$$W_{ij} = \max\left(0, \frac{\vec{v}_i \cdot \hat{d}_{ij}}{\|\vec{v}_i\| + \epsilon}\right) \quad (66)$$

67. Moisture transport edge weight. Atmospheric Transport Graph edge weight.

$$A_{ij}^* = A_{ij} W_{ij} M_{ij} T_{ij} \quad (67)$$

68. Normalized graph convolution. GNN spatial encoding.

$$H^{(\ell+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(\ell)} W^{(\ell)}\right) \quad (68)$$

69. Graph attention score. Learnable neighbour importance.

$$e_{ij} = \text{LeakyReLU}(a^T [W h_i \| W h_j]) \quad (69)$$

70. Graph attention aggregation. Attention-weighted spatial message passing.

$$h'_i = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j\right), \quad \alpha_{ij} = \text{softmax}_j(e_{ij}) \quad (70)$$

8. Regime transition field, temporal smoothing and memory

71. Regime probability change. Detect regime strengthening/weakening.

$$\Delta p_{k,t} = p_{k,t} - p_{k,t-1} \quad (71)$$

72. Transition magnitude. Total movement of the regime probability vector.

$$D_t = \frac{1}{2} \sum_k |p_{k,t} - p_{k,t-1}| \quad (72)$$

73. Exponential smoothing. Reduce probability flicker.

$$\tilde{p}_t = \alpha p_t + (1 - \alpha) \tilde{p}_{t-1} \quad (73)$$

74. EWMA bias memory. Recent forecast-error memory.

$$B_t = \lambda B_{t-1} + (1 - \lambda) e_t \quad (74)$$

75. Grid-regime bias memory. Update regime-specific bias using regime responsibility q .

$$B_{g,r,t} = \lambda B_{g,r,t-1} + (1 - \lambda) e_{g,t} q_{g,r,t} \quad (75)$$

76. Transition bias memory. Remember bias during transitions.

$$B_{g,r \rightarrow s,t} = \lambda B_{g,r \rightarrow s,t-1} + (1 - \lambda) e_{g,t} q_{g,r \rightarrow s,t}^{trans} \quad (76)$$

77. Markov transition probability. Estimate regime-to-regime transition matrix.

$$T_{rs} = P(R_t = s \mid R_{t-1} = r) = \frac{N_{rs}}{\sum_j N_{rj}} \quad (77)$$

78. Transition prediction. One-step regime transition forecast.

$$\hat{p}_{t+1} = \hat{p}_t T \quad (78)$$

79. KL divergence of regimes. Measure distribution shift between consecutive states.

$$D_{KL}(p \parallel q) = \sum_k p_k \log \frac{p_k}{q_k} \quad (79)$$

80. Jensen-Shannon divergence. Symmetric, bounded regime-shift score.

$$JSD(p, q) = \frac{1}{2} D_{KL}(p \parallel m) + \frac{1}{2} D_{KL}(q \parallel m), \quad m = \frac{p + q}{2} \quad (80)$$

9. Mixture of Experts, routing and sparse computation

81. Router logits. Router input from user architecture.

$$z_g = W_r [X_g \parallel P_g \parallel \Delta P_g \parallel S_g \parallel M_g \parallel L_g] + b_r \quad (81)$$

82. Router weights. Expert mixing weights.

$$w_g = \text{softmax}(z_g / T) \quad (82)$$

83. Dense MoE output. All-expert mixture for small K .

$$\hat{y}_g = \sum_{k=1}^K w_{gk} E_k(x_g) \quad (83)$$

84. Top-k mask. Sparse expert selection.

$$m_{gk} = \mathbf{1}\{k \in \text{TopK}(w_g)\} \quad (84)$$

85. Top-k renormalization. Weights sum to one over selected experts.

$$\tilde{w}_{gk} = \frac{m_{gk} w_{gk}}{\sum_j m_{gj} w_{gj} + \epsilon} \quad (85)$$

86. Sparse MoE output. Top-k expert combination.

$$\hat{y}_g = \sum_{k=1}^K \tilde{w}_{gk} m_{gk} E_k(x_g) \quad (86)$$

87. Expert correction. Interpret expert output as direct corrected rainfall.

$$C_k(X_g) = E_k(X_g) - R_g^{NWP} \quad (87)$$

88. Weighted correction form. Regime-weighted bias correction.

$$\hat{R}_g = R_g^{NWP} + \sum_{k \in \text{TopK}} w_{gk} C_k(X_g) \quad (88)$$

89. Load-balancing loss. Encourage balanced expert utilization in sparse MoE.

$$\mathcal{L}_{LB} = \alpha K \sum_{k=1}^K f_k P_k \quad (89)$$

90. Expected expert compute. Shared-backbone compute scaling.

$$\text{Cost} \approx G(C_{shared} + kC_{expert}) \quad (90)$$

10. Bias correction: baselines and adaptive residual correction

91. Additive bias correction. Constant mean-bias baseline.

$$\hat{y}_{corr} = y_{raw} + b \quad (91)$$

92. Multiplicative scaling. Positive-variable baseline.

$$\hat{y}_{corr} = y_{raw} \frac{\bar{y}_{obs}}{\bar{y}_{raw} + \epsilon} \quad (92)$$

93. Local intensity scaling factor. Wet-day precipitation scaling.

$$\alpha = \frac{\bar{y}_{obs, wet}}{\bar{y}_{raw, wet} + \epsilon} \quad (93)$$

94. Residual target. Train ML post-processor on error rather than total rainfall.

$$\Delta y = y_{obs} - y_{NWP} \quad (94)$$

95. Residual model. ML estimate of NWP error.

$$\widehat{\Delta y} = f_{\theta}(X) \quad (95)$$

96. Residual correction. Additive ML bias correction.

$$\hat{y} = y_{NWP} + f_{\theta}(X) \quad (96)$$

97. Nonnegative rainfall output. Guarantee nonnegative prediction.

$$\hat{y}_+ = \text{softplus}(z) = \log(1 + e^z) \quad (97)$$

98. MLP correction head. Small learned head for residual correction, matching the supplied formula note.

$$\widehat{\Delta y} = W_2 \sigma(W_1 F_{conditioned} + b_1) + b_2 \quad (98)$$

99. Grid-weighted loss. Area/reliability weighted model training.

$$\mathcal{L} = \frac{\sum_g w_g \ell_g}{\sum_g w_g} \quad (99)$$

100. Regime-conditioned residual. Regime-aware residual correction.

$$\widehat{\Delta y}_g = \sum_k w_{gk} f_k(X_g, M_g) \quad (100)$$

11. Quantile mapping and distribution-mapping formulas

101. Generic quantile mapping. Core QM transfer function; widely used precipitation post-processing.

$$x_{BC} = F_{obs}^{-1}(F_{mod}(x_{mod})) \quad (101)$$

102. Empirical CDF. Empirical distribution for EQM.

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(x_i \leq x) \quad (102)$$

103. Empirical quantile. Inverse-CDF definition.

$$Q(p) = \inf\{x : F(x) \geq p\} \quad (103)$$

104. Plotting position. Quantile plotting position; choose convention explicitly.

$$p_i = \frac{i - a}{n + 1 - 2a} \quad (104)$$

105. Wet-day probability. Occurrence frequency used in precipitation correction.

$$p_{wet} = \frac{1}{n} \sum_i \mathbf{1}(y_i \geq \tau_{wet}) \quad (105)$$

106. Wet-day threshold matching. Threshold selection to match observed wet-day frequency.

$$\#\{x_{raw} \geq \tau\} = n p_{wet,obs} \quad (106)$$

107. Equidistant CDF matching. EDCDFm-type correction preserving projected change.

$$x_{BC} = x_{fut} + F_{obs,h}^{-1}(F_{fut}(x_{fut})) - F_{mod,h}^{-1}(F_{fut}(x_{fut})) \quad (107)$$

108. Equiratio CDF matching. Multiplicative distribution correction for precipitation.

$$x_{BC} = x_{fut} \frac{F_{obs,h}^{-1}(F_{fut}(x_{fut}))}{F_{mod,h}^{-1}(F_{fut}(x_{fut})) + \epsilon} \quad (108)$$

109. Parametric Q-Q map. Parametric CDF mapping.

$$x_{BC} = F_{obs}^{-1}(F_{mod}(x; \theta_m); \theta_{obs}) \quad (109)$$

110. Quantile-difference diagnostic. Quantify bias by percentile.

$$D_q(p) = Q_{obs}(p) - Q_{mod}(p) \quad (110)$$

12. EMOS, CSG-EMOS, BMA and ensemble statistics

111. Ensemble mean. Primary ensemble predictor.

$$\bar{f} = \frac{1}{K} \sum_{k=1}^K f_k \quad (111)$$

112. Ensemble spread. Predictive dispersion feature.

$$S^2 = \frac{1}{K-1} \sum_{k=1}^K (f_k - \bar{f})^2 \quad (112)$$

113. Gaussian EMOS mean. Canonical Gaussian EMOS location link.

$$\mu = a + b\bar{f} \quad (113)$$

114. Gaussian EMOS variance. Spread-dependent predictive variance.

$$\sigma^2 = c + dS^2, \quad c, d \geq 0 \quad (114)$$

115. Gaussian EMOS density. Predictive distribution.

$$Y \mid f \sim \mathcal{N}(\mu, \sigma^2) \quad (115)$$

116. Gaussian EMOS NLL. Train EMOS parameters by likelihood.

$$\mathcal{L}_{NLL} = \frac{1}{2} \log(2\pi\sigma^2) + \frac{(y - \mu)^2}{2\sigma^2} \quad (116)$$

117. CSG-EMOS PDF. Censored shifted gamma precipitation distribution; shift convention stated explicitly.

$$f_{CSG}(y) = \begin{cases} [1 - G(\delta)]g(y + \delta), & y > 0 \\ G(\delta), & y = 0 \\ 0, & y < 0 \end{cases} \quad (117)$$

118. CSG-EMOS CDF. Probability of precipitation up to y under the same convention.

$$F_{CSG}(y) = \begin{cases} G(y + \delta), & y \geq 0 \\ 0, & y < 0 \end{cases} \quad (118)$$

119. CSG location link. Example linear mean link; fit coefficients on training data.

$$\mu = a_0 + a_1 \bar{f} \quad (119)$$

120. CSG variance link. Heteroscedastic scale link; coefficients fitted from training data.

$$\sigma^2 = b_0 + b_1 S^2 + b_2 \bar{f} \quad (120)$$

13. Loss functions and optimization for TensorFlow/PyTorch

121. MSE loss. Smooth regression baseline.

$$\mathcal{L}_{MSE} = \frac{1}{n} \sum_i (\hat{y}_i - y_i)^2 \quad (121)$$

122. MAE loss. More robust to outliers.

$$\mathcal{L}_{MAE} = \frac{1}{n} \sum_i |\hat{y}_i - y_i| \quad (122)$$

123. Huber loss. Robust residual learning.

$$\mathcal{L}_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \leq \delta \\ \delta(|e| - \frac{1}{2}\delta), & |e| > \delta \end{cases} \quad (123)$$

124. Quantile / pinball loss. Train P10/P50/P90 directly.

$$\rho_\tau(e) = e(\tau - \mathbf{1}[e < 0]) \quad (124)$$

125. Binary cross-entropy. Rain/no-rain or exceedance probability.

$$\mathcal{L}_{BCE} = -[y \log p + (1 - y) \log(1 - p)] \quad (125)$$

126. Focal loss. Emphasize rare heavy/extreme events.

$$\mathcal{L}_{focal} = -\alpha(1 - p_t)^\gamma \log p_t \quad (126)$$

127. Gaussian NLL. Probabilistic regression.

$$\mathcal{L} = \frac{1}{2} \left[\log \sigma^2 + \frac{(y - \mu)^2}{\sigma^2} \right] \quad (127)$$

128. Composite loss. Combine deterministic, probabilistic and regime objectives.

$$\mathcal{L} = \lambda_1 \mathcal{L}_{rain} + \lambda_2 \mathcal{L}_{prob} + \lambda_3 \mathcal{L}_{spatial} + \lambda_4 \mathcal{L}_{regime} \quad (128)$$

129. Gradient descent. Basic optimizer update.

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t) \quad (129)$$

130. Adam update. Common TensorFlow/PyTorch optimizer.

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad \theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (130)$$

14. Probabilistic verification and ensemble scores

131. Brier score. Binary probabilistic rainfall verification.

$$BS = \frac{1}{n} \sum_i (p_i - o_i)^2 \quad (131)$$

132. Brier skill score. Compare with climatology/persistence baseline.

$$BSS = 1 - \frac{BS_{forecast}}{BS_{reference}} \quad (132)$$

133. ROC hit rate. Thresholded probability forecast.

$$HR(t) = \frac{TP(t)}{TP(t) + FN(t)} \quad (133)$$

134. ROC false alarm rate. X-axis of ROC curve.

$$FAR_{roc}(t) = \frac{FP(t)}{FP(t) + TN(t)} \quad (134)$$

135. ROC AUC. Area under ROC curve.

$$AUC = \int_0^1 HR(FAR) d(FAR) \quad (135)$$

136. Ranked probability score. Multi-category probabilistic verification.

$$RPS = \sum_{k=1}^K (F_k - O_k)^2 \quad (136)$$

137. Ranked probability skill score. Skill relative to climatology/reference.

$$RPSS = 1 - \frac{RPS_{forecast}}{RPS_{reference}} \quad (137)$$

138. Continuous ranked probability score. Primary continuous probabilistic score.

$$CRPS(F, y) = \int_{-\infty}^{\infty} [F(z) - \mathbf{1}(z \geq y)]^2 dz \quad (138)$$

139. CRPS ensemble form. Compute CRPS from ensemble samples.

$$CRPS = \frac{1}{K} \sum_m |f_m - y| - \frac{1}{2K^2} \sum_{m=1}^K \sum_{n=1}^K |f_m - f_n| \quad (139)$$

140. CRP skill score. Relative probabilistic skill.

$$CRPSS = 1 - \frac{CRPS_{forecast}}{CRPS_{reference}} \quad (140)$$

15. Categorical rainfall verification: SIH and additional scores

141. Contingency identity. Total cases in binary event verification.

$$N = TP + FP + FN + TN \quad (141)$$

142. Accuracy. Overall classification accuracy.

$$ACC = \frac{TP + TN}{N} \quad (142)$$

143. Frequency bias. Forecast event frequency relative to observed.

$$BIAS = \frac{TP + FP}{TP + FN} \quad (143)$$

144. Probability of detection / hit rate. Fraction of observed events captured.

$$POD = \frac{TP}{TP + FN} \quad (144)$$

145. False alarm ratio. Fraction of forecast events that did not occur.

$$FAR = \frac{FP}{TP + FP} \quad (145)$$

146. Success ratio. Complement of FAR.

$$SR = 1 - FAR = \frac{TP}{TP + FP} \quad (146)$$

147. Critical success index. SIH-required event score.

$$CSI = \frac{TP}{TP + FP + FN} \quad (147)$$

148. Equitable threat score. Threat skill after random hits correction.

$$ETS = \frac{TP - TP_r}{TP + FP + FN - TP_r} \quad (148)$$

149. Random hits. Expected random hits for ETS.

$$TP_r = \frac{(TP + FP)(TP + FN)}{N} \quad (149)$$

150. Heidke skill score. Categorical skill relative to chance.

$$HSS = \frac{2(TP \cdot TN - FP \cdot FN)}{(TP + FN)(FN + TN) + (TP + FP)(FP + TN)} \quad (150)$$

16. Spatial verification, neighbourhood scores and scale analysis

151. Neighbourhood fraction. Forecast fraction above threshold.

$$M_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \mathbf{1}(f_j \geq \tau) \quad (151)$$

152. Observed neighbourhood fraction. Observed exceedance fraction.

$$O_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \mathbf{1}(o_j \geq \tau) \quad (152)$$

153. Fraction Brier score. Unnormalized neighbourhood error.

$$FBS = \frac{1}{N} \sum_i (M_i - O_i)^2 \quad (153)$$

154. Worst fraction Brier score. Normalization for FSS.

$$FBS_{worst} = \frac{1}{N} \sum_i (M_i^2 + O_i^2) \quad (154)$$

155. Fractions skill score. SIH-required spatial rainfall verification.

$$FSS = 1 - \frac{FBS}{FBS_{worst}} \quad (155)$$

156. FSS across times. Aggregate FSS over cases/scales.

$$FSS = 1 - \frac{\sum_{t,i} (M_{ti} - O_{ti})^2}{\sum_{t,i} (M_{ti}^2 + O_{ti}^2)} \quad (156)$$

157. Neighbourhood radius. Convert grid windows to physical scale.

$$r_{phys} \approx r_{grid} \sqrt{(n_x \Delta x)^2 + (n_y \Delta y)^2} \quad (157)$$

158. Mean absolute spatial error. Alternative neighbourhood metric.

$$MASE_{sp} = \frac{1}{N} \sum_i |M_i - O_i| \quad (158)$$

159. Spatial correlation. Spatial pattern agreement.

$$r_{sp} = \frac{\sum_i (F_i - \bar{F})(O_i - \bar{O})}{\sqrt{\sum_i (F_i - \bar{F})^2 \sum_i (O_i - \bar{O})^2}} \quad (159)$$

160. Pattern RMSE. Field-level deterministic spatial error.

$$RMSE_{sp} = \sqrt{\frac{1}{N} \sum_i (F_i - O_i)^2} \quad (160)$$

17. Thresholds, extremes and uncertainty quantification

161. Fixed event indicator. Convert rainfall amount to a categorical event.

$$I_i(\tau) = \mathbf{1}(y_i \geq \tau) \quad (161)$$

162. Climatological percentile threshold. Threshold by local rainfall percentile.

$$\tau_p = F_{clim}^{-1}(p) \quad (162)$$

163. Exceedance probability. Probability of heavy/extreme rainfall.

$$P(Y > \tau) = 1 - F(\tau) \quad (163)$$

164. Empirical exceedance probability. Estimate local event frequency.

$$\hat{P}(Y > \tau) = \frac{1}{n} \sum_i \mathbf{1}(y_i > \tau) \quad (164)$$

165. Prediction interval coverage. Coverage of uncertainty intervals.

$$PICP = \frac{1}{n} \sum_i \mathbf{1}(L_i \leq y_i \leq U_i) \quad (165)$$

166. Mean interval width. Sharpness of interval forecasts.

$$MIW = \frac{1}{n} \sum_i (U_i - L_i) \quad (166)$$

167. Interval score. Penalize wide or missed prediction intervals.

$$IS_\alpha = (U - L) + \frac{2}{\alpha}(L - y)\mathbf{1}(y < L) + \frac{2}{\alpha}(y - U)\mathbf{1}(y > U) \quad (167)$$

168. POT exceedances. Peaks-over-threshold exceedance variable.

$$Z_i = Y_i - u \mid Y_i > u \quad (168)$$

169. GPD PDF. Extreme-value tail model for $z \geq 0$ and valid support.

$$g(z) = \frac{1}{\sigma} \left(1 + \xi \frac{z}{\sigma}\right)^{-1/\xi-1} \quad (169)$$

170. GPD return level. Approximate return level for threshold exceedance rate ζ -u and shape ξ .

$$z_T = u + \frac{\sigma}{\xi} \left[(T \zeta - u)^\xi - 1\right] \quad (170)$$

18. Continual learning, drift detection and adaptive deployment

171. Online SGD update. Per-observation adaptation.

$$\theta_{t+1} = \theta_t - \eta_t \nabla_{\theta} \ell(y_t, \hat{y}_t) \quad (171)$$

172. EWMA error. Smoothed forecast error.

$$E_t = \lambda E_{t-1} + (1 - \lambda) e_t \quad (172)$$

173. EWMA variance. Track error variability.

$$V_t = \lambda V_{t-1} + (1 - \lambda) (e_t - E_t)^2 \quad (173)$$

174. CUSUM statistic. Detect persistent positive error drift.

$$S_t^+ = \max\{0, S_{t-1}^+ + e_t - k\} \quad (174)$$

175. CUSUM negative statistic. Detect persistent negative error drift.

$$S_t^- = \min\{0, S_{t-1}^- + e_t + k\} \quad (175)$$

176. KL data drift. Feature/distribution drift measure.

$$D_{KL}(P_t \| P_{ref}) = \sum_b P_t(b) \log \frac{P_t(b)}{P_{ref}(b)} \quad (176)$$

177. Population stability index. Operational drift diagnostic.

$$PSI = \sum_b (P_t(b) - P_{ref}(b)) \ln \frac{P_t(b)}{P_{ref}(b)} \quad (177)$$

178. Kalman state prediction. Dynamic bias-state prediction.

$$x_t^- = Ax_{t-1} + Bu_t \quad (178)$$

179. Kalman state update. Fuse observation into memory state.

$$x_t = x_t^- + K_t(y_t - Hx_t^-) \quad (179)$$

180. Kalman gain. Optimal linear update gain under Kalman assumptions.

$$K_t = P_t^- H^T (H P_t^- H^T + R)^{-1} \quad (180)$$

19. LoRA, FiLM and regime-conditioned adapter equations

181. Base linear transform. Reference transformation before low-rank adaptation.

$$F = WX \quad (181)$$

182. LoRA low-rank update. Low-rank trainable update.

$$\Delta W = BA \quad (182)$$

183. Adapted weight. Effective weight after LoRA adaptation.

$$W_{adapted} = W + BA \quad (183)$$

184. LoRA forward pass. Equivalent computation for implementation.

$$F_{adapted} = (W + BA)X = WX + B(AX) \quad (184)$$

185. LoRA bottleneck/output update. Compress then restore the update through rank r.

$$Z = AX, \quad \Delta F = BZ = B(AX) \quad (185)$$

186. Scaled LoRA output. Common LoRA scaling form.

$$F_{adapted} = WX + \frac{\alpha}{r}B(AX) \quad (186)$$

187. FiLM scaling and shift. Generate context-dependent modulation parameters.

$$\gamma = G_\gamma(C), \quad \beta = G_\beta(C) \quad (187)$$

188. FiLM transform. Condition features on regime, transition and memory context.

$$F_{FiLM} = \gamma \odot F + \beta \quad (188)$$

189. LoRA + FiLM. Regime-conditioned low-rank adaptation.

$$F_{cond} = \gamma(C) \odot [(W + BA)X] + \beta(C) \quad (189)$$

190. Regime-conditioned MLP correction. Directly connects the supplied LoRA/FiLM idea to a rainfall bias-correction head.

$$\widehat{\Delta y} = W_2 \sigma(W_1 [\gamma(C) \odot ((W + BA)X) + \beta(C)] + b_1) + b_2 \quad (190)$$

20. Deep ensembles, uncertainty decomposition and conformal calibration

191. Deep ensemble mean. Average independent model predictions.

$$\mu = \frac{1}{M} \sum_{m=1}^M \hat{y}_m \quad (191)$$

192. Deep ensemble variance. Epistemic spread proxy across model instances.

$$s^2 = \frac{1}{M-1} \sum_{m=1}^M (\hat{y}_m - \mu)^2 \quad (192)$$

193. Law of total variance. Separate predictive and model-disagreement uncertainty.

$$\text{Var}(Y | X) = \mathbb{E}_M[\text{Var}(Y | X, M)] + \text{Var}_M(\mathbb{E}[Y | X, M]) \quad (193)$$

194. Mixture predictive mean. Mean of an expert predictive mixture.

$$\mu_{mix} = \sum_k w_k \mu_k \quad (194)$$

195. Mixture predictive variance. Within-expert + between-expert uncertainty.

$$\sigma_{mix}^2 = \sum_k w_k (\sigma_k^2 + \mu_k^2) - \mu_{mix}^2 \quad (195)$$

196. Gaussian CRPS closed form. Fast CRPS for Gaussian EMOS outputs.

$$CRPS(\mathcal{N}(\mu, \sigma^2), y) = \sigma \left[z(2\Phi(z) - 1) + 2\phi(z) - \frac{1}{\sqrt{\pi}} \right], \quad z = \frac{y - \mu}{\sigma} \quad (196)$$

197. Conformal absolute residual. Calibration residual for split conformal prediction.

$$r_i = |y_i - \hat{y}_i| \quad (197)$$

198. Conformal quantile. Calibrated residual radius.

$$q_{1-\alpha} = \text{Quantile}_{1-\alpha} \{r_i\}_{i=1}^{n_{cal}} \quad (198)$$

199. Conformal prediction interval. Distribution-free interval under standard split-conformal assumptions.

$$C_{1-\alpha}(x) = [\hat{y}(x) - q_{1-\alpha}, \hat{y}(x) + q_{1-\alpha}] \quad (199)$$

200. Weighted interval score. Evaluate quantile/interval forecasts with a proper-style summary.

$$WIS = \frac{1}{K + \frac{1}{2}} \left(\frac{1}{2} |y - m| + \sum_{k=1}^K \alpha_k^{-1} IS_{\alpha_k} \right) \quad (200)$$

Implementation map: formula -> model component

Layer	Preferred prototype	Key formulas
Grid / geospatial	NumPy + xarray + rasterio	1-10
Preprocessing	NumPy / PyTorch tensors	11-20
Bias diagnostics	NumPy / torch metrics	21-30
Regime discovery	scikit-learn GMM / HDB-SCAN	31-40
Regime classifier	LightGBM / PyTorch MLP	41-50
Probability calibration	softmax + temperature scaling	51-60
Spatial encoder	CNN / GNN / PyTorch Geometric	61-70
Transition + memory	EMA / Markov / Kalman	71-80, 171-180
Sparse MoE	PyTorch custom router	81-90
Bias-correction baseline	scaling + residual ML	91-100
QM	empirical/parametric CDF mapping	101-110
EMOS / CSG-EMOS	scipy + torch / custom likelihood	111-120
Training	PyTorch / TensorFlow	121-130
Probabilistic verification	custom or xskillscore-style metrics	131-140
SIH categorical verification	vectorized metrics	141-150
Spatial verification	neighbourhood kernels	151-160
Extremes / uncertainty	scipy.stats + conformal layer	161-170
Continual learning	online torch + drift monitors	171-180
LoRA / FiLM / adapters	PyTorch custom modules	181-190
Deep ensembles / conformal uncertainty	PyTorch + calibration layer	191-200

Suggested MVP mathematical core

A practical first model can be expressed compactly as

$$\hat{R}_{g,t} = R_{g,t}^{NWP} + \sum_{k \in TopK} \tilde{w}_{gkt} f_k(X_{g,t}, M_{g,t}) + B_{g,t},$$

with

$$\tilde{w}_g = \text{NormalizeTopK}(\text{softmax}(W_r[X_g \| P_g \| \Delta P_g \| S_g \| M_g \| L_g] / T)).$$

Update the memory from verified error using

$$B_{g,t} = \lambda B_{g,t-1} + (1 - \lambda)(Y_{g,t} - \hat{R}_{g,t}).$$

Use QM and CSG-EMOS as strong statistical baselines rather than assuming the neural MoE is automatically better. Evaluate raw NWP, static ML, regime-aware ML, MoE, and memory-augmented MoE under the same time-blocked verification periods.

Verification design recommendations

- For deterministic rainfall: RMSE, MAE, bias, correlation and grid-weighted scores.
- For event forecasts: POD, FAR, CSI and ETS at multiple precipitation thresholds.
- For spatial structure: FSS at multiple neighbourhood sizes and thresholds.
- For probability forecasts: Brier score, Brier skill score, ROC/AUC, reliability, CRPS and CRPSS.
- For extremes: threshold exceedance probability, interval coverage, interval width and a tail model such as GPD when justified.

- Always stratify verification by lead time, season, regime, grid/region and rainfall threshold; otherwise a national aggregate can hide systematic regime-specific failures.

Key references and formula sources

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Formula conventions. Unless stated otherwise, sums are over valid training/verification samples; epsilon is a small positive constant used only for numerical stability; probability vectors are normalized to sum to one; and all learned parameters are fitted on a training period that precedes the verification period.